CS201 Lab 9

1. **Coupled oscillators:** The equations for two oscillators coupled by a spring are given by,

$$m_1\ddot{x_1} = -(k_1 + k_2)x_1 + k_2x_2,$$
$$m_2\ddot{x_2} = k_2x_1 - (k_2 + k_3)x_2.$$

   Solve the equations numerically and plot x_1 and x_2 (on separate graphs) against t for each of the following conditions:

   (a) $m_1 = m_2 = 1$ and $k_1 = k_2 = k_3 = 1$.

   (b) $m_1 = 1, m_2 = 200$ and $k_1 = k_2 = k_3$. Interpret the result physically.

   (c) $m_1 = m_2 = 1$ and $k_1 = k_3 = 1, k_2 = 0.1$. You should be able to observe the phenomenon of beats for this choice of parameters.

2. **Nonlinear oscillator and some differences with the linear case:** Consider the nonlinear pendulum (without small angle approximation),

$$\ddot{x} + \sin(x) = 0,\ x(0) = A,\ \dot{x}(0) = 0.$$

   (a) Plot t vs $x(t)$ for $A = 1.0$ for the above system and compare with the results of the usual linear pendulum for the same initial conditions (plot both nonlinear and linear results on the same graph for comparison).

   (b) Investigate whether the time period depends on the initial amplitude A for the nonlinear pendulum. Consider three values of $A = 10, 15, 20$ and print the values of the time period for each. Note that any dependence of the time period on initial amplitude marks a difference with the linear pendulum, for which the time period should not depend on A ($T_{\text{linear}} = 2\pi\sqrt{\frac{L}{g}}$).
   (Think of a way to make your code calculate the time period.)